A full volume theorem under complete-radius balance

Original source and complete geometry

Fixed certified simple-white plat; prescribed knot-producing parity.
Exact shared-face coefficient, marked zeros, both level parities.

$$\lim_{N\to\infty} \frac{\log Z_{D,N}}{N} \quad = \quad \max_{q\in\mathcal{Q}_D} \sum_e h(q_e) \quad = \quad \frac{\mathrm{Vol}(Y_D)}{2\pi}$$

Entire CROSS filling period: proved

Nonzero geometric saddle; all original Fourier charges and
whole-domain endpoint currents controlled.

$$\lim_{N\to\infty} \frac{2\pi}{N} \log |I_{\mathrm{cross},N}| \quad = \quad \mathrm{Vol}(S^3 \setminus K_D(\mathbf{p}))$$

Actual balance at the same complete packing

$$Q(x,b,y) + Q(x,c,y) > \log t \quad (t \in \{b,c\},\ t > 0)$$

One sufficient test, with $A = x + y > 0$, is

$$(A+b)(A+c) \geq \max\{b^2, c^2\}.$$

The explicit family also uses the sharper zero-middle test $e/A \leq 9/4$.

Full original invariant: proved under that condition

The LEFT/RIGHT remainder has a strict global exponential gap.

$$\lim_{N\to\infty} \frac{2\pi}{N} \log |K_N(K_D(\mathbf{p}))| \quad = \quad \mathrm{Vol}(S^3 \setminus K_D(\mathbf{p}))$$

Every independently long fixed signed twist vector;
threshold depends on D. Twists are fixed before $N \to \infty$.

Proved family at EVERY fixed bridge width $k \geq 2$

Three original layers, $3k-3$ independently long signed twist regions.
A uniform packing recurrence verifies the balance hypothesis.

Exponents divisible by 3 give minimal bridge number exactly k.

Open: extend to general plat diagrams

The finite balance theorem applies beyond this family whenever its test holds.
A general estimate when balance fails remains to be proved.